

Multiple Choice (10 marks)

1. Let $x_1 = 1$ and $x_{n+1} = \sqrt{3+2x_n}$ for all positive integers n . If it is assumed that $\{x_n\}$ converges, then $\lim x_n =$
 - (a) 3
 - (b) e
 - (c) $\sqrt{5}$
 - (d) 0
2. If $c > 0$ and $f(x) = e^x - cx$ for all real numbers x , then the minimum value of f is
 - (a) $f(c)$
 - (b) $f(e^c)$
 - (c) $f(1/c)$
 - (d) $f(\log c)$
3. The set of integers $\mathbb{Z}$ with the binary operation $*$ defined as $a*b = a + b + 1$ for a, b in $\mathbb{Z}$, is a group. The identity element of this group is
 - (a) 0
 - (b) 1
 - (c) -1
 - (d) 12
4. Let $y = f(x)$ be a solution of the differential equation $x dy + (y - xe^x) dx = 0$ such that $y = 0$ when $x = 1$. What is the value of $f(2)$?
 - (a) $1/(2e)$
 - (b) $1/e$
 - (c) $e^2/2$
 - (d) $2e$
5. Statement 1 | Any set of two vectors in $\mathbb{R}^2$ is linearly independent. Statement 2 | If $V = \text{span}(v_1, \dots, v_k)$ and $\{v_1, \dots, v_k\}$ are linearly independent, then $\dim(V) = k$.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: There is only one additive abelian group of order 16 that satisfies the condition $x + x + x + x = 0$ for all x in the group.
7. True or False: The maximum possible order for an element of S_6 is 12.
8. True or False: The complex numbers under addition form a group, including the zero element as the identity.
9. True or False: The norm $\|x\|_p$ can be expressed as an inner product only for $p = 2$.
10. True or False: The largest order of an element in the group of permutations of 5 objects is 6.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. On her previous five attempts Sarah had achieved times, in seconds, of 86, 94, 97, 88 and 96, for swimming 50 meters. After her sixth try she brought her median time down to 92 seconds. What was her time, in seconds, for her sixth attempt?
12. Find the distance between the points (0,15) and (8,0).

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